

Laravel topics covered

Basics of Laravel

- What Laravel is and why it is used
- Basic folder structure (routes, app, database, etc.)
- Running a Laravel project
- Using .env file for configuration
- Basic use of Artisan commands

Database Setup

- Connecting Laravel with a database
- Setting database name, username, password in .env
- Running migrations to create tables
- Understanding tables and columns

Migrations

- Creating migration files
- Adding columns like name, email, password
- Running migrations using Artisan
- Rolling back migrations

Routes

Web Routes

- Creating routes in web.php
- Returning simple responses (text/views)

API Routes

- Creating routes in api.php
- Returning JSON data

Controllers

- Creating controllers using Artisan
- Moving logic from routes to controllers
- Creating simple functions (methods) inside controllers
- Returning responses from controllers

Requests & Validation

- Getting data from user input
- Validating input (required fields, email, etc.)
- Showing validation errors

Authentication

Using Laravel Sanctum

- Basic login and registration system
- Generating tokens for users
- Protecting routes using authentication

Session Basics

- Understanding login sessions
- Keeping users logged in

Middleware

- What middleware is
- Protecting routes (only logged-in users can access)
- Using built-in auth middleware

File Handling

- Uploading files (images, etc.)
- Saving files in storage
- Accessing uploaded files

Storage

- Using Laravel storage system
- Linking storage folder (storage:link)
- Accessing files via browser

Pagination

- Showing data in pages (not all at once)
- Using simple pagination

Filtering Data (Backend)

- Filtering data based on conditions (e.g., user_id, category)
- Using query parameters (e.g., ?status=active)
- Returning filtered results from the database

Infinite Scroll (Basic)

- Loading more data when scrolling
- Connecting frontend with backend for more data

API Basics

- Creating simple APIs
- Sending JSON responses
- Using basic status codes (200, 404, 422)

CORS

- Basic idea of CORS
- Allowing frontend to connect with backend

Blade Frontend

- Creating frontend pages using Blade templates
- Passing data from controllers to Blade views
- Using Blade directives (@if, @foreach, etc.)
- Layouts and reusable components

Inertia.js Frontend

- Connecting Laravel with React using Inertia
- Passing backend data directly to frontend pages
- Building SPA-like applications without separate API setup

PHP String Functions Learned

- `strtoupper()` → Convert text to uppercase
- `strtolower()` → Convert text to lowercase
- `ucwords()` → Capitalize first letter of each word
- `strlen()` → Get string length
- `substr()` → Extract part of a string
- `explode()` → Convert string into array
- `implode()` → Convert array into string